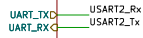


PowerDist

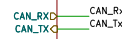
File: PowerDist.kicad_sch

USB



File: USB.kicad_sch

Car_CAN



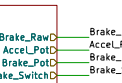
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Motor_CAN



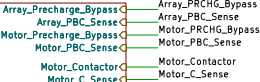
File: Motor_CAN.kicad_sch

Potentiometers



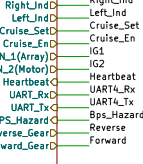
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Contactors



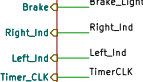
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DashAndDisplay

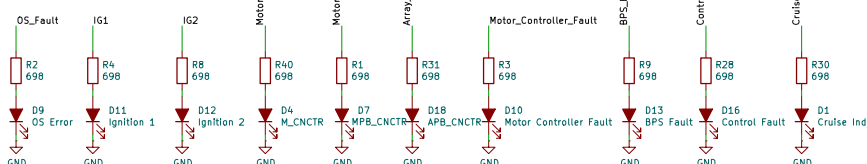
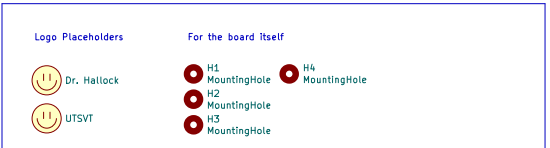
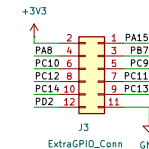
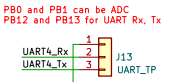
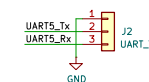
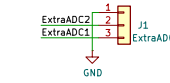
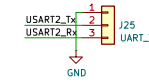
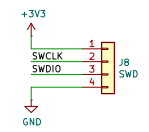
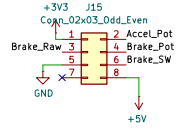
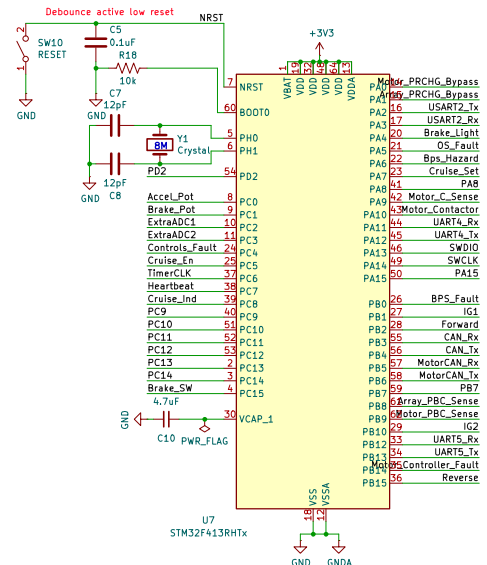
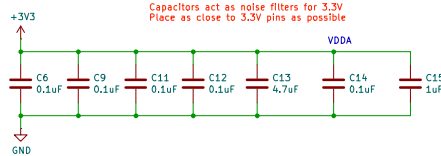


File: MinionBrdInterface.kicad_sch

Lights

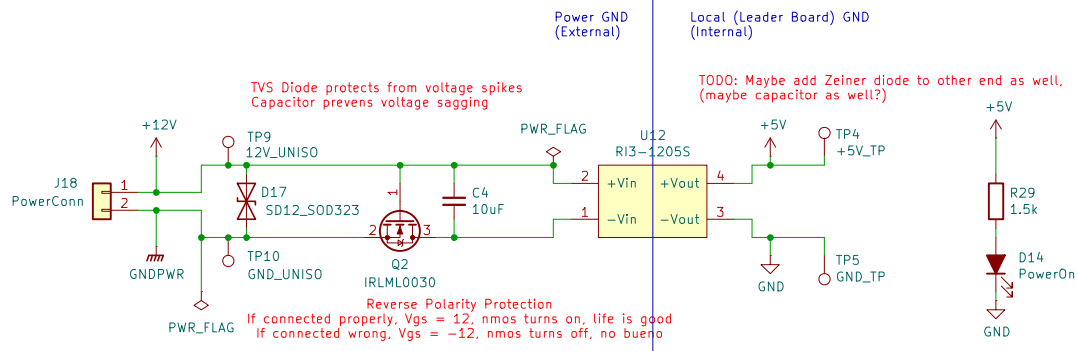


File: Lights.kicad_sch

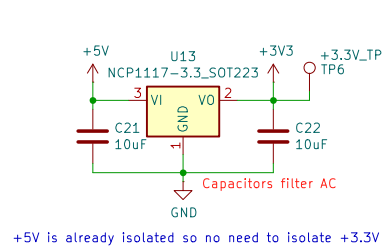


Controls Leaderboards: Daybreak Rework 2025 - Updated for assembly order
 - An intermediate board between the 2022-2024 and 2024-2026 cars that merges the Daybreak Controls Leaderboard and Lighting Board.
 - Created in conjunction with the new 2025 Controls Dashboard.
 - See sub-sheets for additional revisions since last version.
 Authors: Krish Parikh, Madeleine Lee
 LHRs
 Sheet: /
 File: Controls-Leader.kicad_sch
Title: Controls Leader
 Size: A3 Date: 2025-01-18 Rev: 2.5
 KiCad E.D.A. 8.0.5 Id: 1/10

ISOLATED

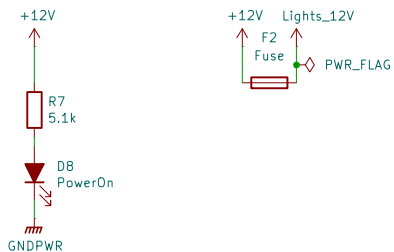


Isolated DC-DC Converter to step down +12V power to an isolated +5V. The input gnd (Power GND) is different from the output gnd (Local GND)



Linear Regulator to step down +5V to +3.3V.
An isolated converter was not used for the +3.3V line because of price and space. May need to change this to DC-DC converter if we really want to make the BPS more energy efficient.

Linear Regulator is about 50 cents
DC-DC converter would be 11 dollars
Doesn't seem like it would be a worthwhile investment, unless we have leftover budget that we have nothing to spend on



- Removed jumper and distinction between GNDPWR/GND1, +12V/+12VA
- Added fuse for lights (+12V to Lights_12V)

Sheet: /PowerDist/
File: PowerDist.kicad_sch

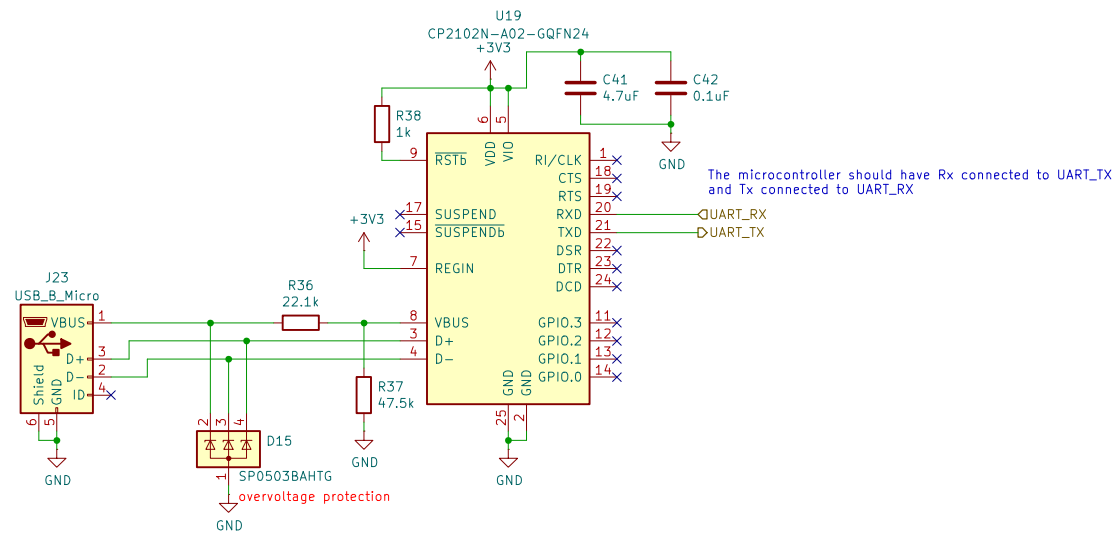
Title:

Size: A4
KiCad E.D.A. 8.0.5

Date:

Rev:

Id: 2/10



- Updated USB chip

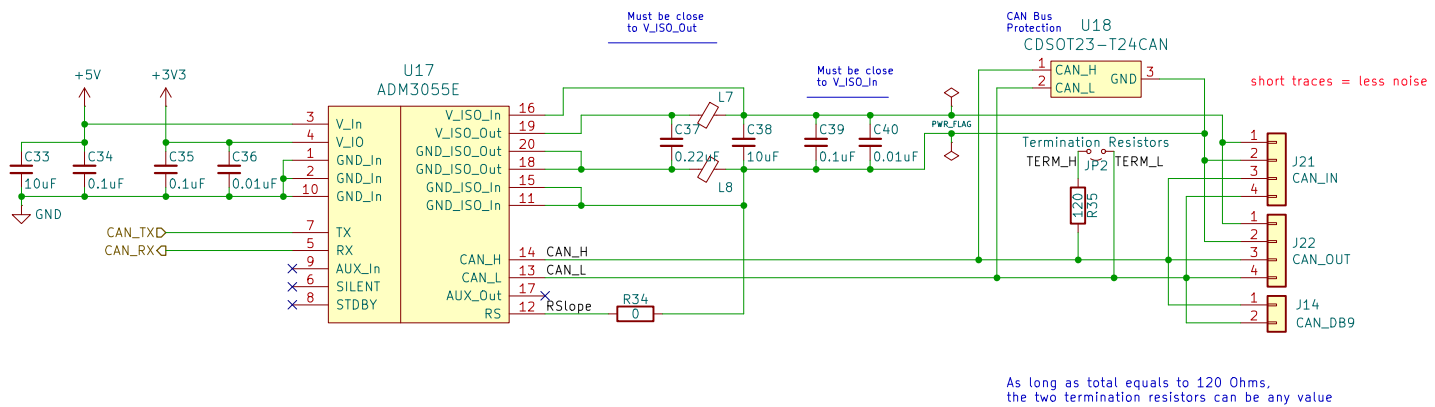
Sheet: /USB/
File: USB.kicad_sch

Title:

Size: A4
KiCad E.D.A. 8.0.5

Date:

Rev:
Id: 5/10



- Added J14 CAN_DB9 for debugging
- Removed testpoint
- Chose new ferrite beads

Sheet: /Car_CAN/
File: CAN.kicad_sch

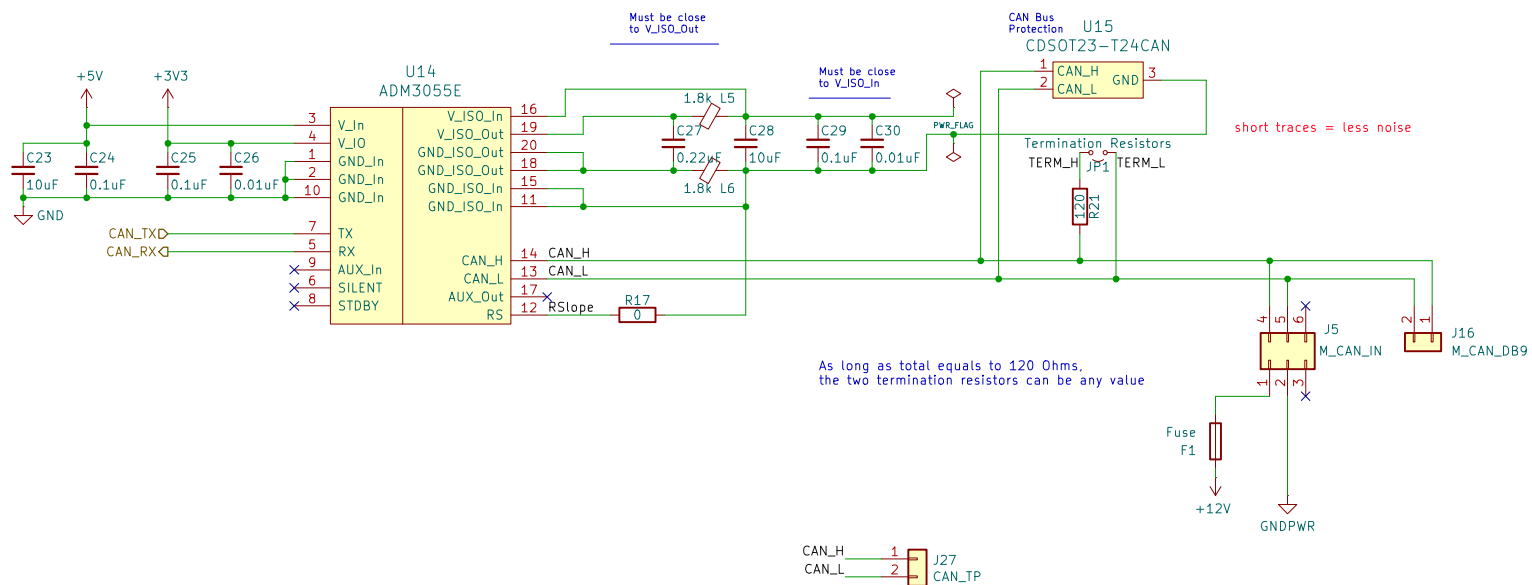
Title:

Size: A4
KiCad E.D.A. 8.0.5

Date:

Rev:

Id: 6/10



- Changed motorcan connector from big round to molex
- Added 375mA fuse
- Added J16 M_CAN_DB9
- Chose new ferrite beads

Sheet: /Motor_CAN/
File: Motor_CAN.kicad_sch

Title:

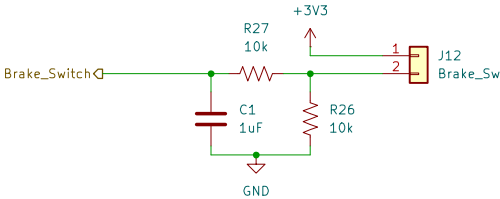
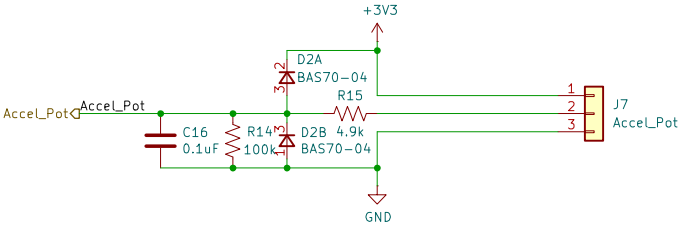
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KiCad E.D.A. 8.0.5

Date:

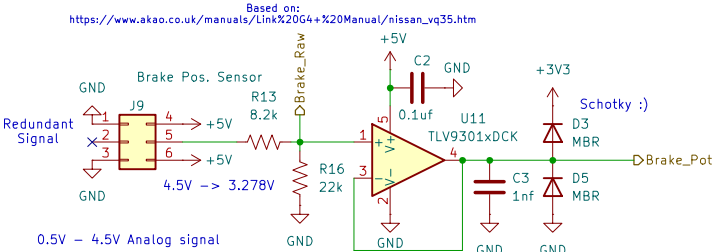
Rev:

Id: 7/10

Potentiometers



Brake Pedal



- Updated accel pot values to increase ADC range
- Added brake switch button circuit

Sheet: /Potentiometers/
File: Potentiometers.kicad_sch

Title:

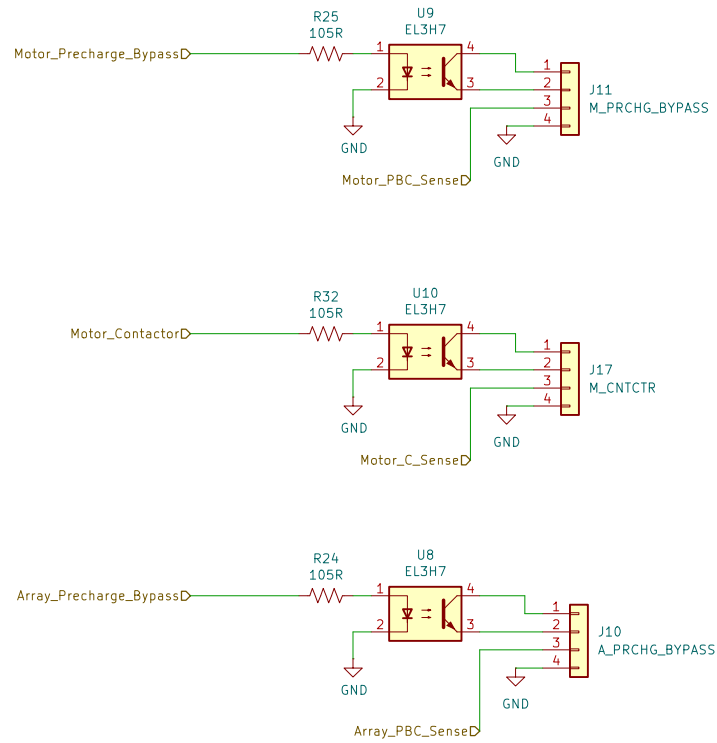
Size: A4
KiCad E.D.A. 8.0.5

Date:

Rev:

Id: 8/10

Contactors



- Separated contactors to 1 per connector
- Added sense pins
- Removed array contactor

Sheet: /Contactors/
File: Contactors.kicad_sch

Title:

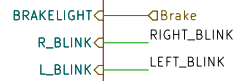
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KiCad E.D.A. 8.0.5

Date:

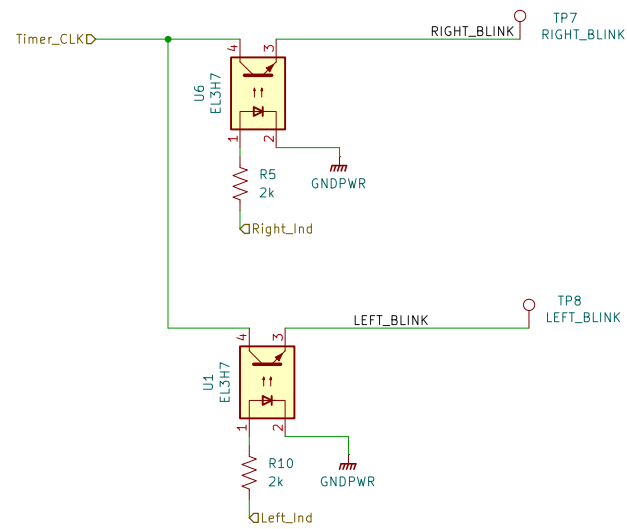
Rev:

Id: 9/10

Lights Drivers



File: Lights_Drivers.kicad_sch



- Replaced lights drivers/circuits
- Replaced 555 timer with blinking gpio that passes through to drivers when dashboard Left/Right 3-pos switch is toggled
- Intended to connect to breakout boards that split out the necessary number of lights

Sheet: /Lights/
File: Lights.kicad_sch

Title:

Size: A4
KiCad E.D.A. 8.0.5

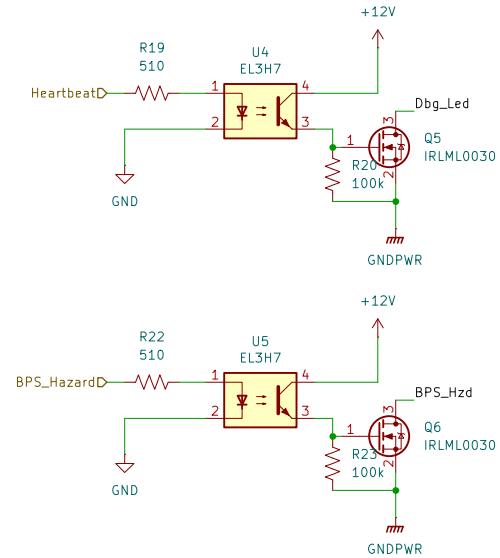
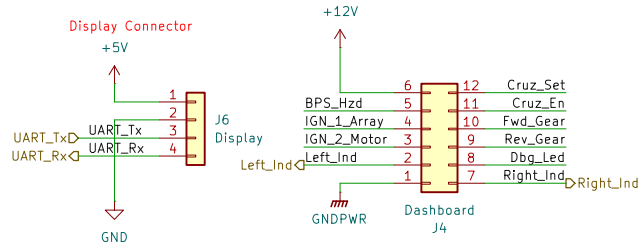
Date:

Rev:

Id: 9/10

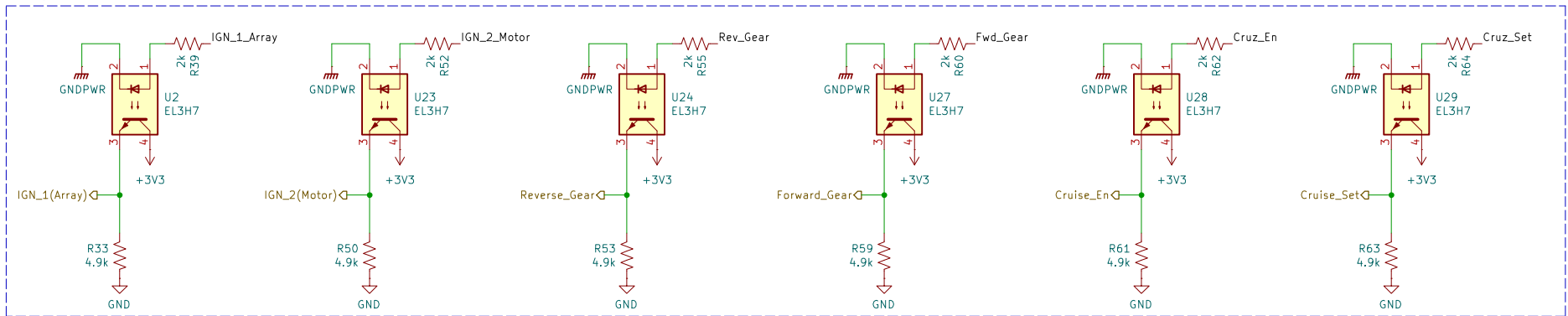


Connectors + Optoisolators



Optoisolator has abs max If of 60mA. This 2k r should give a current of ~6mA.

Inputs from dashboard



- Renamed pins
- Exporting to and receiving from dashboard as 12V
- Added heartbeat LED, BPS_Hazard since Controls is now handling BPS faults

Sheet: /DashAndDisplay/
File: MinionBrdInterface.kicad_sch

Title:

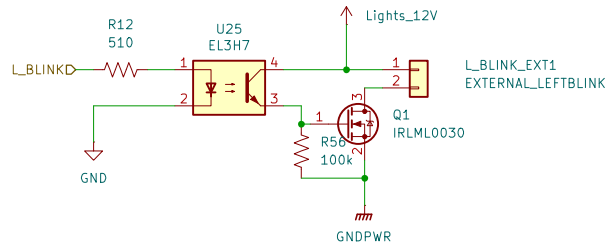
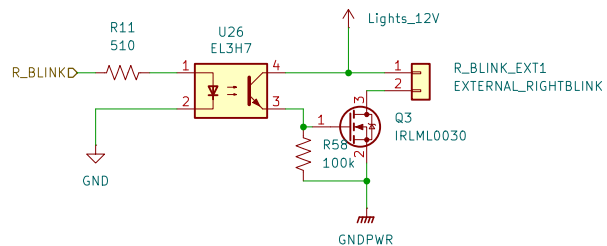
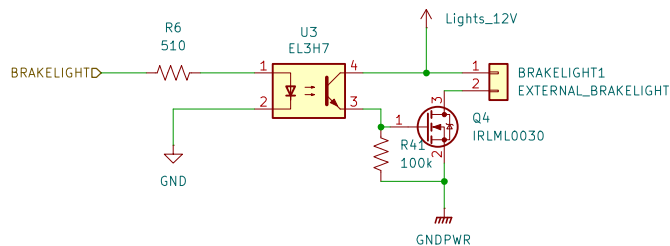
Size: A4 Date:

KiCad E.D.A. 8.0.5

Rev:

Id: 10/10

Assuming that all external
lights have I-Lim Res



Sheet: /Lights/Lights Drivers/
File: Lights_Drivers.kicad_sch

Title:

Size: A4 Date:

KiCad E.D.A. 8.0.5

Rev:

Id: 10/10